

FRD – Customer Supply Chain Management Dashboard

1. Project Objective

Develop an interactive Executive Business Intelligence Dashboard to enable stakeholders to monitor Sales, Customer, Product, and Logistics performance across global markets. The dashboard should provide real-time insights into operational efficiency, profitability, customer behavior, and supply chain performance.

2. Business Problem

The organization generates large volumes of transactional supply chain data, making it difficult to extract meaningful insights through raw tables alone. Stakeholders lack a centralized reporting solution to monitor sales trends, shipping performance, customer segmentation, and product profitability. This limits their ability to identify performance issues, optimize logistics, and improve overall business efficiency.

3. Dashboard Sections

The report consists of **4 interactive pages**.

3.1 Executive Overview

Purpose

Provide an overall business performance summary.

KPIs

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Total Quantity Sold
- Late Delivery %

Visuals

- Sales Trend
- Profit Trend
- Sales by Product
- Orders by Category
- Sales by Market

3.2 Customer Analytics

Purpose

Analyze customer behavior, profitability, and segmentation.

KPIs

- Total Customers
- Average Revenue per Customer
- Average Order Value
- Average Profit per Customer

Visuals

- Sales by Customer Segment
- Top Customers by Revenue
- Customer Distribution Map

- Monthly Customer Sales Trend
- Customer Profitability Scatter Plot
- Customers by State

3.3 Product & Sales Analytics

Purpose

Evaluate product performance and profitability.

KPIs

- Total Products
- Total Revenue
- Total Profit
- Quantity Sold
- Average Product Price
- Average Discount

Visuals

- Top 10 Products
- Bottom 10 Products
- Revenue by Department
- Product Profitability
- Profit by Category
- Price Range Analysis

3.4 Shipping & Logistics Analysis

Purpose

Monitor delivery performance and logistics efficiency.

KPIs

- On-time Delivery %

- Late Delivery %
- Average Shipping Days
- Average Delay
- Total Shipments
- Cancelled Shipments

Visuals

- Delivery Status Trend
- On-time Delivery by Shipping Mode
- Late Deliveries by Market
- Average Shipping Days by Region
- Delivery Performance Matrix
- Scheduled vs Actual Shipping Days

4. Data Requirements

FACT TABLE

- Fact_order_item (fact table)
- Order Item Id
- Customer_id
- Order Id
- Geo_key
- Product Card Id
- Shipping_key
- Date_key
- Benefit per order
- Order Item Discount
- Order Item Discount Rate
- Order Profit Per Order
- Sales per customer
- Order Item Product Price

- Order Item Profit Ratio
- Order Item Quantity
- Sales
- Order Item Total

DIMENSION TABLE

- dim_shipping
- dim_product
- dim_department
- dim_category
- dim_geography
- dim_customers
- dim_date
- state_ab

5. Filters

Global slicers available on all pages.

- Year
- Month
- Market
- Category
- Customer Segment

Cross-filtering enabled across visuals.

Interactivity

- **Custom Tooltip:** Hovering over the **Sales Trend** visual displays a contextual business summary for the selected date, including key performance indicators such as Sales, Profit, Orders, Customers, Profit Margin, Average Discount, Late Delivery %, and On-time Delivery %. The tooltip updates dynamically based on applied filters and the selected point in the trend.

- **Drill-through:** Users can select any product from the Product & Sales dashboard to open a dedicated Product Detail page displaying product attributes and overall business performance.

6. Visual Requirements

Page	Visual	Purpose
Executive	KPI Cards	Business summary
Executive	Line Charts	Sales & Profit trend
Executive	Bar Charts	Product, Category & Market comparison
Customer	Donut Chart	Customer segment contribution
Customer	Map	Customer distribution
Customer	Scatter Plot	Customer profitability
Customer	Bar Chart	Top customers
Product	Horizontal Bars	Top & Bottom products
Product	Scatter Plot	Discount vs Profit
Product	Department Chart	Revenue contribution
Product	Category Chart	Category performance

Shipping	Line Chart	Delivery status trend
Shipping	Column Chart	Shipping mode comparison
Shipping	Scatter Plot	Shipping performance
Shipping	Clustered Column	Scheduled vs Actual days

7.MEASURES

Sales Measures

- Total Sales

Total Sales = SUM(fact_order_item[Order Item Total])

- Total Profit

Total profit = SUM(fact_order_item[Order Profit Per Order])

- Profit Margin %

Profit margin % = DIVIDE([Total profit],[Total Sales])

- Total Orders

Total orders = DISTINCTCOUNT(fact_order_item[Order Item Id])

- Total Quantity Sold

Total Quantity Sold = SUM(fact_order_item[Order Item Quantity])

Customer Measures

- Total Customers

Total Customers = DISTINCTCOUNT(fact_order_item[customer_id])

- Average Revenue per Customer

AVG REVENUE/CUSTOMER = DIVIDE([Total Sales],DISTINCTCOUNT(fact_order_item[customer_id]))

- Average Order Value

AVG ORDER VALUE = DIVIDE([Total Sales],[Total orders])

- Average Profit per Customer

AVG PROFIT/CUSTOMER = DIVIDE([Total profit],[Total Customers])

Product Measures

- Total Products

Total Products = DISTINCTCOUNT(fact_order_item[Product Card Id])

- Average Product Price

Average Product Price = AVERAGE(fact_order_item[Order Item Product Price])

- Average Discount

Average Discount = AVERAGE(fact_order_item[Order Item Discount Rate])

- Product Profit Ratio

profit margin % = DIVIDE([Total profit],[Total Sales])

Logistics Measures

- Average Shipping Days

AVG SHIP DAYS =

**AVERAGEX(fact_order_item,

RELATED(dim_shipping[Days for shipping (real)])

)**

- Average Delay

AVG DELAY = AVERAGE(dim_shipping[difference])

- Late Delivery %

**Late delivery % = DIVIDE(CALCULATE(COUNTROWS(fact_order_item),

dim_shipping[Delivery Status]="Late delivery"),

[Total orders],0)**

- On-time Delivery %

**On-time delivery % =
DIVIDE(COUNTROWS(FILTER(fact_order_item,RELATED(dim_shipping[Delivery
Status])= "Shipping on time")),[Total orders])**

- Cancelled Shipments

**Cancelled Shipments = CALCULATE(DISTINCTCOUNT(fact_order_item[Benefit per
order]),dim_shipping[Delivery Status]="Shipping canceled")**

Time Intelligence

- Previous Year Sales
- Previous Year Profit
- Year-over-Year Growth
- Monthly Sales Trend
- Monthly Profit Trend

8.Export / Reporting Needs

- Export Options: PDF, Excel, Power BI Service shareable link
- Scheduled Refresh: Daily at 6 AM
- Distribution: Internal analytics team, management dashboard

9.Notes / Special Instructions

- Null or invalid data handled.
- Ensure all visuals maintain consistency in color and format across pages.